# Vanishing Activity in Exact Causal Simulation

Causal Embedding, an Infinite-Horizon Trichotomy, and the Entropy of the Last Event

Jan Mikulík

Companion note to `finite-Shannon-theorem`

October 2026

**Abstract**

For a time-inhomogeneous binary controlled process with maximal conditional success probabilities $p_t \le 1/2$, we give a single, policy-independent, countably valued coordinate construction and establish its exact entropy cost. This yields a three-way classification of exact common causal seeds: absence of any countable seed, existence only at infinite entropy, and existence at finite entropy. In the critical regime $p_t = 1/[t(\ln t)^2]$, the finite-horizon cost is $\log_2 \ln n + O(1)$. A strengthening shows that the last success time of just one policy already carries $\log_2 \ln n + \kappa + o(1)$ bits. A power-logarithmic family exhibits a broader phase diagram. The arguments here are mathematical proofs, not claims that their full Lean formalization is already present in the repository.

## 1 Model and causal embedding

At time $t \ge 3$, the available action set $A_t(h)$ is finite and nonempty at every finite binary history $h$. The action $a$ gives a Bernoulli output with

$$0 \le q_t(h,a) := \mathbb{P}(Y_t = 1 \mid h, a) \le p_t \le \tfrac{1}{2}, \qquad p_t > 0. \tag{1}$$

For every history, some action attains $q_t(h,a) = p_t$. A *common causal seed* is a random element $S$ and a deterministic function specifying the output at every history and action, such that each deterministic adaptive policy has precisely its stipulated transcript law. The seed and response functions must be fixed before the policy is selected. Let $C_n$ be the infimum of Shannon entropy in bits among seeds exact through time $n$.

**Theorem 1** (Exact causal embedding and finite-horizon sandwich)**.** *Under* (1)*, for every* $n \ge 3$*,*

$$\boxed{\sum_{t=3}^{n} h_2(p_t) \;\le\; C_n \;\le\; \sum_{t=3}^{n} h_2(p_t) + 2\sum_{t=3}^{n} p_t.} \tag{2}$$

*The upper bound is attained by an explicit countable product seed; for any finite horizon its readout can be represented by a finite seed without increasing entropy.*

*Proof.* Choose mutually independent random variables $Z_t \in \mathbb{N}_0$ with

$$\mathbb{P}(Z_t = 0) = 1 - p_t, \qquad \mathbb{P}(Z_t = k) = p_t 2^{-k} \quad (k \ge 1).$$

For every $(t,h,a)$ select digits $b_k(t,h,a) \in \{0,1\}$ such that

$$q_t(h,a)/p_t = \sum_{k \ge 1} b_k(t,h,a) 2^{-k}.$$

This includes 0 (all zero digits) and 1 (all one digits). Define the strategy determined by $S = (Z_t)_{t \ge 3}$ by

$$D_t(S;h,a) = \begin{cases} 0, & Z_t = 0, \\ b_{Z_t}(t,h,a), & Z_t \ge 1. \end{cases}$$

Every length-$(t-1)$ history is measurable with respect to $(Z_3, \dots, Z_{t-1})$; the policy's selected action is likewise past-measurable. Independence of $Z_t$ therefore gives, at every reached history,

$$\mathbb{P}\big(D_t(S; h, a) = 1 \mid \text{past}\big) = \sum_{k \geq 1} p_t 2^{-k} b_k(t, h, a) = q_t(h, a).$$

Induction over time gives the product formula for every controlled transcript; in particular one and the same decoder is exact for all adaptive policies. The positive geometric law $2^{-k}$ has two bits of entropy, so the entropy chain rule gives

$$H(Z_t) = h_2(p_t) + 2p_t, \qquad H(Z_3, \dots, Z_n) = \sum_{t=3}^{n} [h_2(p_t) + 2p_t].$$

At horizon $n$, each seed value induces one total binary response strategy on a finite tree. Pushing the countable law forward onto the finite strategy set leaves all transcript laws unchanged and cannot increase entropy.

For the lower bound, choose at each history an action attaining $p_t$. Under this deterministic policy the outputs $B_t$ are independent Bernoulli($p_t$) variables. Their joint transcript is a deterministic image of any exact seed, hence $H(S) \geq H(B_3, \dots, B_n) = \sum_{t=3}^{n} h_2(p_t)$. Take the infimum over exact seeds. □

# 2 Three infinite-horizon regimes

**Theorem 2** (Countability and Shannon trichotomy)**.** *For an exact common seed on all times $t \geq 3$, precisely one of the following holds:*

1. *If $\sum_t p_t = \infty$, no countably supported exact common seed exists.*
2. *If $\sum_t p_t < \infty$ and $\sum_t h_2(p_t) = \infty$, a countably supported exact common seed exists, but all such seeds have infinite entropy.*
3. *If $\sum_t h_2(p_t) < \infty$, a countably supported exact common seed of finite entropy exists.*

*Proof.* Under the maximizing policy from Theorem 1, each fixed infinite binary transcript $(b_t)$ has probability at most

$$\prod_{t \geq 3} (1 - p_t),$$

using $p_t \leq 1/2$. This product vanishes if $\sum_t p_t = \infty$. The maximizing transcript law is then non-atomic, whereas a deterministic image of a countably supported law is atomic. This proves (1).

If $\sum_t p_t < \infty$, the first Borel–Cantelli lemma gives $\mathbb{P}(Z_t \neq 0 \text{ infinitely often}) = 0$. The set of sequences with finitely many nonzero coordinates, each a positive integer, is countable; the construction in Theorem 1 is therefore a countably supported seed on the whole infinite tree. Exactness holds at every finite horizon. Its entropy equals $\sum_t [h_2(p_t) + 2p_t]$, possibly infinite. The lower bound of (2) shows that any exact seed has infinite entropy when $\sum_t h_2(p_t) = \infty$. Finally $h_2(p) \geq p$ on $(0, 1/2]$, so summability of $h_2(p_t)$ implies summability of $p_t$, and the explicit seed has finite entropy. This proves (2)–(3). □

**Remark 1** (An alternative bridge to atomic capacity)**.** *Suppose the existing infinite-tree atomic-capacity theorem is applicable to the local kernels (in particular, its strict-positivity assumptions, when required, are met). For any policy and $n > N$, the probability of a 1 after $N$ is at most $\sum_{t>N} p_t$. Thus at least $1 - \sum_{t>N} p_t$ of the transcript mass is carried by at most $2^{N-2}$ transcripts, those that are all zero after $N$. Consequently*

$$\text{atomQ} \;\geq\; 1 - \sum_{t>N} p_t.$$

*If $\sum_t p_t < \infty$, then* $\text{atomQ} = 1$. *Conversely, if $\sum_t p_t = \infty$, the maximizing policy has every individual length-$n$ transcript of probability at most $\prod_{t=3}^{n}(1 - p_t)$. Therefore, for any fixed $K$, its $K$ heaviest transcripts carry mass at most $K \prod_{t=3}^{n}(1 - p_t) \to 0$. Taking the infimum over horizons and policies gives* $\text{atomQ} = 0$. *Thus in this class the atomic capacity obeys the exact* zero–one law

$$\boxed{\text{atomQ} = \begin{cases} 1, & \sum_t p_t < \infty, \\ 0, & \sum_t p_t = \infty. \end{cases}}$$

*The repository's atomic-capacity/countability equivalence therefore supplies an* alternative *existence proof. This route does not give the explicit entropy upper bound, and zero-probability branches must be handled using the appropriate zero-kernel model.*

# 3 Critical logarithmic regime and the last event

Take $p_t = 1/[t(\ln t)^2]$, $t \geq 3$.

**Theorem 3** (Double-logarithmic cost and asymptotic sufficiency of the last event)**.** *Let $B_t$ be the maximizing policy's independent* Bernoulli$(p_t)$ *outputs, and put $J_n = \max\big(\{t \in \{3,\dots,n\} : B_t = 1\} \cup \{0\}\big)$. Then*

$$C_n = \log_2 \ln n + O(1), \tag{3}$$

$$H(J_n) = \log_2 \ln n + \kappa + o(1) \tag{4}$$

*for some finite constant $\kappa$. More precisely,*

$$\boxed{\sum_{t=3}^{n} h_2(p_t) - H(J_n) = \sum_{s=3}^{n-1} h_2(p_s)\Big(1 - \prod_{t=s+1}^{n} (1-p_t)\Big).} \tag{5}$$

*The right-hand side converges to a finite limit as $n \to \infty$. The infinite last-success time $J = \max(\{t \geq 3 : B_t = 1\} \cup \{0\})$ is almost surely finite but has $H(J) = \infty$ and $\mathbb{E}J = \infty$.*

*Proof.* As $p \downarrow 0$,

$$h_2(p) = \frac{p\ln(1/p) + p + O(p^2)}{\ln 2}.$$

Hence

$$h_2(p_t) = \frac{1}{\ln 2}\frac{1}{t\ln t} + O\left(\frac{1+\ln\ln t}{t(\ln t)^2}\right).$$

The error is summable, and the sum–integral discrepancy for $1/(t\ln t)$ converges. Therefore $\sum_{t=3}^{n} h_2(p_t) = \log_2 \ln n + c_h + o(1)$ for a finite $c_h$. Since $\sum_t p_t < \infty$, (3) follows from (2).

Condition on $J_n = t > 0$. Then $B_t = 1$ and $B_{t+1},\dots,B_n = 0$, while $(B_3,\dots,B_{t-1})$ retain their original independent distribution. On $J_n = 0$, all entries are zero. The chain rule and the fact that $J_n$ is a function of $(B_3,\dots,B_n)$ give

$$H(B_3,\dots,B_n) - H(J_n) = H(B_3,\dots,B_n \mid J_n) = \sum_{s=3}^{n-1} h_2(p_s)\mathbb{P}(J_n > s),$$

which is (5). By the union bound,

$$0 \leq \mathbb{P}(J_n > s) \leq \sum_{t>s} p_t = O((\ln s)^{-1}).$$

Thus $\sum_s h_2(p_s)\sup_{n>s}\mathbb{P}(J_n > s)$ is dominated by a constant multiple of $\sum_s 1/[s(\ln s)^2] < \infty$. Dominated convergence in (5) gives a finite limit $K_\infty$, whence (4) with $\kappa = c_h - K_\infty$.

By Borel–Cantelli, $J < \infty$ almost surely. For $t \geq 3$,

$$\mathbb{P}(J = t) = p_t\prod_{u>t}(1-p_u) \sim p_t, \quad \mathbb{P}(J \geq n) = 1 - \prod_{t\geq n}(1-p_t) \sim \sum_{t\geq n} p_t \sim \frac{1}{\ln n}.$$

Consequently $H(J) = \infty$ by comparison with $\sum_t p_t \log_2(1/p_t)$, and $\mathbb{E}J = \sum_{n\geq 1}\mathbb{P}(J \geq n) = \infty$. □

**Proposition 1** (The last event detects the entropy branch)**.** *For any sequence $0 < p_t \leq 1/2$ with $\sum_t p_t < \infty$, let $J$ be the last success of independent* Bernoulli$(p_t)$ *outputs. Then*

$$\boxed{H(J) < \infty \iff \sum_t h_2(p_t) < \infty.}$$

*Proof.* Write $r_t = \mathbb{P}(J = t) = p_t R_t$, where $R_t = \prod_{u>t}(1-p_u) \to 1$. Eventually $p_t < e^{-1}$ and $1/2 \le R_t \le 1$. Therefore

$$\tfrac{1}{2} p_t \log_2(1/p_t) \le -r_t \log_2 r_t \le -p_t \log_2 p_t$$

for all sufficiently large $t$, by monotonicity of $-x \log_2 x$ on $(0, e^{-1})$. The remaining terms are finite, and $\sum_t h_2(p_t) < \infty$ holds exactly when $\sum_t p_t \log_2(1/p_t) < \infty$, since $\sum_t p_t < \infty$. The atom at $J = 0$ contributes only a finite term. □

**Remark 2** (What "vanishing" does and does not mean)**.** *For the explicit common seed,* $T = \max(\{t : Z_t \neq 0\} \cup \{0\}) < \infty$ *almost surely. For each fixed realization of that seed,* every policy simultaneously *outputs zero after* $T$*. Nevertheless* $\mathbb{P}(T \ge n) \sim 1/\ln n$ *and* $\mathbb{E}T = \infty$*. The finite number of active times has finite expectation,* $\mathbb{E}\sum_t \mathbf{1}_{\{Z_t \neq 0\}} = \sum_t p_t < \infty$*, while the last active* time *need not have finite expectation.*

# 4 Power-logarithmic phase diagram

**Corollary 1** (A family of thresholds)**.** *For* $\alpha > 1$ *let* $p_t = 1/[t(\ln t)^\alpha]$ *(*$t \ge 3$*). An exact countably supported causal seed exists, and*

$$C_n = \begin{cases} \dfrac{(\ln n)^{2-\alpha}}{(2-\alpha)\ln 2} + O(1), & 1 < \alpha < 2, \\ \log_2 \ln n + O(1), & \alpha = 2, \\ C_{\lim} + o(1), \quad C_{\lim} < \infty, & \alpha > 2. \end{cases}$$

*For* $0 \le \alpha \le 1$*,* $\sum_t p_t = \infty$*, so no exact countably supported infinite-horizon seed exists.*

*Proof.* For $\alpha > 1$, $\sum_t p_t < \infty$ and the error in

$$h_2(p_t) = \frac{1}{\ln 2\, t(\ln t)^{\alpha-1}} + O\left(\frac{1 + \ln\ln t}{t(\ln t)^\alpha}\right)$$

is summable. Integral comparison yields the first two cases and convergence of $\sum_t h_2(p_t)$ exactly when $\alpha > 2$. Theorem 1 bounds the excess uniformly in $n$. When $\alpha > 2$, the monotone finite-horizon costs have a finite limit $C_{\lim}$ by the bounds. The repository's projective identity identifies this limit with the infinite-horizon optimum when its hypotheses apply. Theorem 2 gives all countability statements. □

# 5 Lean integration targets

The updated repository already verifies the geometric entropy identity, $H(Z_t) = h_2(p_t) + 2p_t$, finite excess, and $h_2(p) \ge p$ in `CausalSeed/Vanishing.lean`. The next proof obligations are separate:

1. **Causal embedding:** define the dyadic-digit readout on the existing infinite-tree strategies; prove the conditional Bernoulli law at every history and hence the path-mass identity.
2. **Countability:** prove the finite-support sequence lemma under $\sum p_t < \infty$ and obtain a ranked, countable law on strategies, or alternatively use `atomQ` and a uniform tail bound where its hypotheses apply.
3. **Entropy:** connect the coordinate product bound and maximizing-policy data processing to `Completion.finiteCost`; transfer the upper bound through finite-strategy pushforward.
4. **Analysis:** formalize the summable remainder in the binary-entropy expansion and the integral-test asymptotics. The last-event identity is a useful independent benchmark lemma.

No kernel-checked claim is made here for these four bridges until the corresponding Lean declarations compile and pass an axiom audit.

# References


[1] J. Mikulík, *One Seed for Every Adaptive Policy*, `https://github.com/jendamikulik/finite-Shannon-theorem` (repository, accessed October 2026).